Enrollment No._____

GANPAT UNIVERSITY

M.C.A. SEMESTER-I EXAMINATION

Nov-DEC 2022

P11A1JP : JAVA PROGRAMMING

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following **(8)**

(A) Define : Data Abstraction , Polymorphism

(B) How super keyword and this keyword works?

2. Answer the following (Any Two) **(11)**

(A) What is class and object? Give example of constructor overloading with suitable example.

(B) Explain Java primitive Datatypes.

(C) What is Bytecode? Explain use of static and instance methods with example and state how both differ with each other.

3. Answer the following (Any Two) **(11)**

(A) What is Abstract class? Give example of abstract class with exception handling.

(B) What is Package? Give example of package and class with various access modifiers.

(C) Types of Java Inheritance with example

SECTION-II

4. Answer the following **(8)**

(A) Types of Data structure.

(B) Define : Tree ,Graph

5. Answer the following (Any Two) **(11)**

(A) How stack is used for factorial algorithm using Recursion.

(B) Explain Bubble sort algorithm with example.

(C) How binary search and sequential search differ with each other? Explain sequential search algorithm with example.

6. Answer the following (Any Two) **(11)**

(A) Explain stack operations algorithms with example.

(B) What is Circular Queue? Explain algorithm of Circular Queue.

(C) Write an algorithm to perform insertion operation of single linked list.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.C.A. SEMESTER-I EXAMINATION NOV-DEC 2022

P11A2PP PYTHON PROGRAMMING

Time: 3 Hours] **[Total Marks: 60**

Instructions to the examinee:

1. Figures to the right indicate full marks.
2. Each section should be written in a separate answer book.
3. Be precise and to the point in your answer.

SECTION-I

1. Answer the following. **(08)**

1. List out any six features of Python.
2. What is an Array? Describe its types.
3. How Date & Time will display in Python? Explain.
4. Write a difference between Break and Continue.

2. Answer the following (ANY Two) **(11)**

1. Write a short note on Data types of Python.
2. What is Recursive function? Elaborate with suitable example.
3. Explain Function and its types in Python.

3. Answer the following (ANY Two) **(11)**

1. How lambda function works? Write a small program to multiply two numbers using lambda function.
2. What is Flow Control? Explain any three statements.
3. What is String? Describe Six methods with suitable example.

SECTION-II

4. Answer the definition with suitable example. **(08)**

1. Class and Object
2. Histogram
3. Data frame
4. Constructor

5. Answer the following (ANY Two) **(11)**

1. How to handle exception? Describe with example.
2. How to convert (1) CSV (2) Dictionary (3) List into Data frame with suitable example.
3. What is Inheritance? Elaborate all its types.

6. Answer the following (ANY Two) **(11)**

1. What is Matplotlib? Draw a Line graph with line width as 20.5 and values are [3, 8, 1, 10].
2. What is Chart? Creating Pie Charts with labels such as "Apples", "Bananas", "Cherries", "Dates" and its values as 35, 25, 25, 15 respectively.
3. Write a different operation on Frame with suitable example.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.C.A. SEMESTER-I EXAMINATION

NOV-DEC 2022

P11A3DMS: DATABASE MANAGEMENT SYSTEM

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following **(08)**

1. What is Functional dependencies? Explain it in detail.
2. Discuss 1NF and 2NF With Example.

2. [A] Answer the following (ANY TWO) **(10)**

1. Explain Query Processing with diagram.
2. Write a note on DBMS and SQL performance tuning in detail.
3. Write a short note on Database Query Optimization Modes.

[B] What is Normalization? **(01)**

3. [A] Answer the following (ANY TWO) **(10)**

1. What is Deadlock? Discuss Deadlock Prevention in detail.
2. Discuss Concurrency control with Locking Methods.
3. Explain Timestamp-Based Protocols in Concurrency.

[B] What is Concurrency control? **(01)**

(P.T.O)

SECTION-II

4. Answer the following **(08)**

1. Give the difference between procedure and function in PL/SQL.
2. Explain pre-defined exception handling with example.

5. [A] Answer the following (ANY TWO) **(10)**

1. What is PL/SQL? Write down the block structure of PL/SQL.
2. Explain IF …ELSE…. with example.
3. What is cursor? Discuss the types of cursor.

[B] What is full form of DDL? **(01)**

6. [A] Answer the following (ANY TWO) **(10)**

1. What is Trigger? Explain it with proper example
2. Discuss the use of procedure with suitable example.
3. Explain how to Create, Invoke and Alter Package in PL/SQL?

[B] What is Package? **(01)**

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.C.A. SEMESTER-I EXAMINATION (NOV-DEC 2022)

P11A4FM : Foundations of Mathematics

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following **(8)**

(A) If $y = 5\cos x - 3\sin x$, prove that $\frac{d2y}{dx2}+y=0$

(B) In a housing society, half of the families have a single child per family, while the remaining half has two children per family. Find the probability that a child picked at random, has a sibling.

2. Answer the following (Any Two) **(11)**

(A) Prove that $(A \cap B)' = A' \cup B'$.(De Morgan's theorem)

(B) If ${}^{n-1}P_3 : {}^{n}P_4 = 1 : 6$, then what is the value of n?

(C) In a group of 70 people, 45 people speak Hindi and 33 speak English language and 10 people speak neither English nor Hindi. How many people speak both? How many speak only English? How many speak only Hindi?

3. Answer the following (Any Two) **(11)**

(A) Find the stationary points, maximum value and minimum value of the curve given by $f(x) = 2x^3 - 9x^2 + 12x - 3$ in the two dimensional Cartesian plane and in the closed interval $0 \le x \le 3$.

(B) Prove that if A and B are sets, then $P(A \times B) = P(A) \times P(B)$ where P stands for the power set.

(C) A man and his wife appear in an interview for two vacancies in the same post. The probability of husband's selection is 1/7 and the probability of wife's selection is 1/5. What is the probability that only one of them is selected?

SECTION-II

4. **Answer the following** **(8)**

(A) The average age of a class of 39 students is 15 years. If the age of a teacher is included, the average increases by 3 months. Find the age of teacher.

(B) Ram got 30% in a test and failed by 10 marks. If the pass marks in the test was 70, find the maximum (full) marks in it.

5. **Answer the following (Any Two)** **(11)**

(A) If three numbers are added in pairs, the sums equal 10,19and 21. Find those three numbers.

(B) Raju was 4 times as old as his son 8 years ago. After 8 years, Raju will be twice as old as his son. What are their ages?

(C) A person incurs 5% loss by selling a watch for Rs.1140. At what price should the watch be sold to earn 5% profit?

6. **Answer the following (Any Two)** **(11)**

(A) A,B and C form a partnership, A invests 3 times as much as B invests and B invests two-third of what C invests. At the end of the year, the profit earned is Rs. 6600. What are the shares of A, B and C?

(B) A man travelled in a car from his village to the post-office at the rate of 25 kmph and walked back at the rate of 4 kmph. If the whole journey took 5 hours 48 minutes, find the average speed, the total distance and actual distance between the post-office.

(C) A sum of Rs. 1550 is lent out into two parts, one at 8% and another part at 6%. If the total annual income is Rs.106. Find the two parts of the amount at different rates.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

M.C.A. SEMESTER-I EXAMINATION

NOV-DEC 2022

P11A5LFN: LINUX FUNDAMENTAL WITH NETWORKING

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following. **(8)**

(1) Explain the `awk` filter.

(2) Explain purpose, syntax and usage of the Linux commands: `cut` and `paste`.

2. Answer the following. (ANY TWO) **(11)**

(1) Draw and explain Linux architecture.

(2) Write a C program to copy a text file to another file.

(3) Explain the Linux commands: `wc` and `sort`

3. Answer the following. (ANY TWO) **(11)**

(1) Explain working of the system calls: `readdir`, `mkdir`, `rmdir`

(2) Explain purpose, syntax, switches of the Linux `grep` command using suitable examples.

(3) What is purpose of the Linux commands `pipe` and `tee`? Using suitable example explain their usage.

SECTION-II

4. Answer the following. **(8)**

(1) What is role of signal in Linux? Explain any two signals.

(2) Explain usage of: `msgget` and `msgsnd`

5. Answer the following. (ANY TWO) **(11)**

(1) What is zombie process? Explain purpose, syntax and usage of `wait` and `waitpid` in Linux.

(2) Explain purpose, syntax and usage of `fork` and `vfork`.

(3) How do the Linux system calls `execl` and `execv` differ?

6. Answer the following. (ANY TWO) **(11)**

(1) Using suitable example, explain purpose and usage of the `pipe` system call in Linux.

(2) Explain purpose and usage of `semget`, `semop` and `semctl` in Linux.

(3) Explain purpose and usage of `socket`, `sendmsg` and `recvmsg` in Linux.

END OF PAPER